



EDUCORE CAMPUS OS

Shadow Segment Role Ownership

Three shadow segments, seven balanced roles, and exhaustive ownership for the simplified, fresher-friendly project.

3 Shadow segments	7 Engineering roles	155 Files assigned	0 Missing or duplicate
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Restructured edition - 02 July 2026. Tests and historical one-off maintenance scripts are removed. backend/app/database/seed.py is the only seed file and contains the complete working dataset.

Coverage rule: Every project-owned source or documentation file is assigned exactly once. Local dependencies, virtual environments, caches, generated builds, output PDFs, runtime logs, uploads, and backend/.env secrets are excluded.

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1. Restructured ownership outcome

EduCore keeps conventional modular layers while removing the pieces that made the training project feel historical or temporary. The backend now has one supported CLI, one authoritative seed file, no project test layer, no committed runtime logs, and no one-off seed, reseed, fix, verify, cleanup, or migration helper scripts. Useful project guidance is grouped under docs/.

3	7	155	0
Shadow segments	Engineering roles	Files assigned	Missing or duplicate

Shadow segment	Role split	Files	Files per role
Frontend	Role 1 + Role 2	44	22 + 22
Database	Role 3 + Role 4	44	22 + 22
Backend	Role 5 + Role 6 + Role 7	67	22 + 22 + 23
Total	7 roles	155	Range 22-23

Balance result: six roles own 22 files and one role owns 23. The difference between the lightest and heaviest role is one file.

Inventory boundary

- Includes 155 project-owned source, configuration, migration, documentation, template, and brand files.
- Includes all seven organized docs/ guides because they support setup, training, presentation, and role clarity.
- Excludes .git, node_modules, virtual environments, Python caches, frontend/dist, output PDFs, tmp files, runtime logs, upload contents, and backend/.env.
- The ownership PDF is excluded from its own inventory because it is generated output.

2. Shadow segment charters

Frontend - 44 files

Browser experience, reusable interface platform, product workflows, frontend delivery, and user-facing project guidance. **Roles 1 and 2, with 22 files each.**

Database - 44 files

Persisted data meaning, ORM entities, core validation contracts, migrations, repositories, sessions, and the complete seed. **Roles 3 and 4, with 22 files each.**

Backend - 67 files

HTTP services, security, academic and learning rules, finance and library operations, runtime behavior, and operational guidance. **Roles 5 and 6 own 22 files each; Role 7 owns 23.**

3. Seven-role overview

Role	Segment	Title	Files
1	Frontend	Frontend Platform, Design System and Delivery Engineer	22
2	Frontend	Frontend Workflows, Contracts and Product Experience Engineer	22
3	Database	Database Schema, Models and Data Contracts Engineer	22
4	Database	Database Migrations, Repositories and Reliability Engineer	22
5	Backend	Backend Platform, Security and Cross-Domain API Engineer	22
6	Backend	Academic, People and Learning Backend Engineer	22
7	Backend	Finance, Library and Operational Backend Engineer	23

Primary ownership versus shared review

- Primary ownership means accountability for implementation quality, manual checks, documentation, review routing, and readiness.
- A primary owner may not unilaterally change persisted meaning, a public API, a permission boundary, financial state, or production operation.
- Database changes require Database review; user-facing changes require Frontend review; transport, security, and domain behavior require the relevant Backend role.
- Ownership follows responsibility rather than folder name when a contract, template, or guide primarily serves another segment.

4. Feature-to-segment ownership

Feature area	Lead segment	Lead role	Mandatory supporting review
Application shell, shared UI, frontend delivery	Frontend	Role 1	Role 5 for session/security; Role 7 for deployment
Pages, forms, role workspaces, product guidance	Frontend	Role 2	Relevant Backend owner; Role 3 for data meaning
ORM entities, constraints, core contracts	Database	Role 3	Role 4 migration; affected Backend and Frontend owners
Migrations, repositories, sessions, seed	Database	Role 4	Role 3 plus every affected domain owner
Authentication, RBAC, API assembly, dashboard	Backend	Role 5	Frontend roles and Database roles
Academics, people, learning, timetable, reports	Backend	Role 6	Role 2, Role 3, Role 4, and Role 5
Finance, library, runtime operations	Backend	Role 7	Role 2, Role 3/4, and Role 5

5. Cross-segment change rules

Change	Primary owner	Required reviewers
Reusable component or navigation behavior	Role 1	Role 2; Role 5 if auth or permission related
Page, form, or role-specific workflow	Role 2	Role 1 plus relevant Backend owner and Role 3
Field, table, relationship, or constraint	Role 3	Role 4 plus affected Backend and Frontend owners
Migration, seed, or data repair	Role 4	Role 3 plus every affected domain owner
Endpoint, permission, token, or middleware	Role 5	Frontend and relevant Database/domain owners
Academic, attendance, marks, assignment, timetable rule	Role 6	Roles 2, 3, 4, and Role 5 for scope
Payment, salary, circulation, runtime, deployment	Role 7	Roles 2, 3/4, and Role 5 for security/audit

6. Frontend shadow segment

This segment contains 44 files across 2 accountable roles.

Role	Title	Files
1	Frontend Platform, Design System and Delivery Engineer	22
2	Frontend Workflows, Contracts and Product Experience Engineer	22

Role 1 - Frontend Platform, Design System and Delivery Engineer

Shadow segment: Frontend Exact file load: 22

Own the browser foundation, shared components, session integration, global styling, dependencies, build tooling, reverse proxy, and frontend delivery.

Deep responsibility scope

- React bootstrapping, protected routes, session state, API requests, and permission-aware navigation.
- Reusable layout and UI components, global responsive styles, loading states, and shared interaction behavior.
- Vite, ESLint, package management, Nginx, containers, branding, and production frontend output.

Files owned by this role

113. frontend/.dockerignore

Purpose: Keeps local dependencies, caches, and generated output out of container build contexts.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

114. frontend/Dockerfile

Purpose: Builds the service into a repeatable production container image.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

115. frontend/eslint.config.js

Purpose: Implements frontend behavior for eslint.config.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

116. frontend/index.html

Purpose: Defines the HTML structure for index.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

117. frontend/nginx.conf

Purpose: Supports the project behavior represented by nginx.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

118. frontend/package-lock.json

Purpose: Pins the complete frontend dependency graph for reproducible installs.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

119. frontend/package.json

Purpose: Declares frontend scripts and direct dependencies.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

120. frontend/public/institute.png

Purpose: Provides the EduCore institute brand image used by the application and documentation.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

121. frontend/src/App.jsx

Purpose: Defines protected routes, lazy page loading, role gates, HOD scope, and route fallbacks.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

122. frontend/src/components/layout/AppShell.jsx

Purpose: Provides responsive navigation, permission-aware resources, workspace search, notifications, account controls, and sign-out.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

123. frontend/src/components/layout/PermissionGate.jsx

Purpose: Provides the reusable Permissiongate browser component.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

124. frontend/src/components/ui/Button.jsx

Purpose: Provides the reusable Button browser component.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

125. frontend/src/components/ui/Card.jsx

Purpose: Provides the reusable Card browser component.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

126. frontend/src/components/ui/ConfirmDialog.jsx

Purpose: Provides the reusable Confirmdialog browser component.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

127. frontend/src/components/ui/PageHeader.jsx

Purpose: Provides the reusable Pageheader browser component.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

131. frontend/src/components/ui/StatCard.jsx

Purpose: Provides the reusable Statcard browser component.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

135. frontend/src/lib/api.js

Purpose: Centralizes browser API requests, credentials, error parsing, downloads, and authentication failure handling.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

136. frontend/src/main.jsx

Purpose: Implements frontend behavior for main.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

151. frontend/src/state/AuthContext.jsx

Purpose: Maintains the signed-in user, session loading, login, logout, and permission-aware browser state.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

152. frontend/src/styles/dashboard-widgets.css

Purpose: Defines shared browser presentation rules for dashboard widgets.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

153. frontend/src/styles/index.css

Purpose: Defines shared browser presentation rules for index.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

155. frontend/vite.config.js

Purpose: Implements frontend behavior for vite.config.

Feature contribution: Contributes to a consistent, secure, responsive browser platform and repeatable frontend delivery.

What this role may edit/change: May change shared browser architecture, components, styling, build, or delivery behavior; run lint/build and manually verify affected responsive and permission states.

Required review/dependencies: Role 2 reviews user impact; Role 5 reviews session/security; the relevant Backend role reviews API behavior.

Role 2 - Frontend Workflows, Contracts and Product Experience Engineer

Shadow segment: Frontend Exact file load: 22

Own the user-facing pages, forms, resource definitions, people workflows, feature styling, and the main project guide.

Deep responsibility scope

- Role-specific pages for academics, attendance, assignments, finance, timetables, profiles, and notifications.
- Form definitions, resource metadata, validation feedback, loading, empty, success, and failure experiences.
- Product documentation and shared people/profile/form experiences that connect browser flows to backend contracts.

Files owned by this role

002. README.md

Purpose: Explains the product, capabilities, simplified structure, startup flow, seed commands, and verification steps.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

128. frontend/src/components/ui/PeopleDirectory.jsx

Purpose: Provides the reusable Peopledirectory browser component.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

129. frontend/src/components/ui/ProfileDetailModal.jsx

Purpose: Provides the reusable Profiledetailmodal browser component.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

130. frontend/src/components/ui/RecordFormModal.jsx

Purpose: Provides the reusable Recordformmodal browser component.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

132. frontend/src/config/resourceForms.js

Purpose: Declares reusable create and edit form fields, validation choices, and feature-specific inputs.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

133. frontend/src/config/resources.jsx

Purpose: Declares reusable resource tables, columns, endpoints, permissions, and display behavior.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

134. frontend/src/config/roleAccess.js

Purpose: Maps roles and permissions to browser-visible workspaces and navigation behavior.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

137. frontend/src/pages/AssignmentsPage.css

Purpose: Styles the assignmentspage workflow across desktop and responsive layouts.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

138. frontend/src/pages/AssignmentsPage.jsx

Purpose: Implements the browser workflow and role-aware experience for assignmentspage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

139. frontend/src/pages/AttendancePage.css

Purpose: Styles the attendancepage workflow across desktop and responsive layouts.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

140. frontend/src/pages/AttendancePage.jsx

Purpose: Implements the browser workflow and role-aware experience for attendancepage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

141. frontend/src/pages/DashboardPage.jsx

Purpose: Implements the browser workflow and role-aware experience for dashboardpage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

142. frontend/src/pages/FeeCollectionPage.css

Purpose: Styles the feecollectionpage workflow across desktop and responsive layouts.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

143. frontend/src/pages/FeeCollectionPage.jsx

Purpose: Implements the browser workflow and role-aware experience for feecollectionpage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

144. frontend/src/pages/HodManagementPage.jsx

Purpose: Implements the browser workflow and role-aware experience for hodmanagementpage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

145. frontend/src/pages/HodPanelPage.jsx

Purpose: Implements the browser workflow and role-aware experience for hodpanelpage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

146. frontend/src/pages/LoginPage.jsx

Purpose: Implements the browser workflow and role-aware experience for loginpage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

147. frontend/src/pages/NotificationsPage.jsx

Purpose: Implements the browser workflow and role-aware experience for notificationspage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

148. frontend/src/pages/ProfilePage.jsx

Purpose: Implements the browser workflow and role-aware experience for profilepage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

149. frontend/src/pages/ResourcePage.jsx

Purpose: Implements the browser workflow and role-aware experience for resourcepage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

150. frontend/src/pages/TimetablePage.jsx

Purpose: Implements the browser workflow and role-aware experience for timetablepage.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

154. frontend/src/styles/people-directory.css

Purpose: Defines shared browser presentation rules for people directory.

Feature contribution: Contributes to clear, role-appropriate workflows that expose every supported feature without duplicating business rules.

What this role may edit/change: May change user-facing pages, forms, resource metadata, and product copy; verify validation, loading, empty, success, and failure states with the relevant API owner.

Required review/dependencies: Role 1 reviews shared UI; the relevant Backend role reviews workflow behavior; Role 3 reviews data-contract meaning.

7. Database shadow segment

This segment contains 44 files across 2 accountable roles.

Role	Title	Files
3	Database Schema, Models and Data Contracts Engineer	22
4	Database Migrations, Repositories and Reliability Engineer	22

Role 3 - Database Schema, Models and Data Contracts Engineer

Shadow segment: Database Exact file load: 22

Own persisted data meaning, SQLAlchemy entities, relationships, constraints, and the core academic and people validation contracts.

Deep responsibility scope

- ORM tables, relationships, enumerations, soft deletion, identifiers, and cross-entity integrity.
- Academic, attendance, course, department, and student request/response contracts.
- Compatibility analysis for every field, relationship, constraint, and sensitive-data exposure decision.

Files owned by this role

035. backend/app/database/base.py

Purpose: Defines the declarative ORM base and shared model behavior used by every persisted entity.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

041. backend/app/models/__init__.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for init.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

042. backend/app/models/academic.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for academic.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

043. backend/app/models/attendance.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for attendance.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

044. backend/app/models/audit_log.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for audit log.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

045. backend/app/models/course.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for course.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

046. backend/app/models/department.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for department.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

047. backend/app/models/finance.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for finance.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

048. backend/app/models/library.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for library.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

049. backend/app/models/marks.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for marks.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

050. backend/app/models/notification.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for notification.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

051. backend/app/models/student.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for student.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

052. backend/app/models/subject.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for subject.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

053. backend/app/models/teacher.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for teacher.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

054. backend/app/models/timetable.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for timetable.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

055. backend/app/models/timetable_grid.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for timetable grid.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

056. backend/app/models/user.py

Purpose: Defines SQLAlchemy persistence entities, relationships, and constraints for user.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

070. backend/app/schemas/academic.py

Purpose: Defines Pydantic request and response contracts for academic.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

072. backend/app/schemas/attendance.py

Purpose: Defines Pydantic request and response contracts for attendance.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

075. backend/app/schemas/course.py

Purpose: Defines Pydantic request and response contracts for course.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

077. backend/app/schemas/department.py

Purpose: Defines Pydantic request and response contracts for department.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

081. backend/app/schemas/student.py

Purpose: Defines Pydantic request and response contracts for student.

Feature contribution: Contributes to stable persisted meaning, relational integrity, and explicit core data contracts.

What this role may edit/change: May change fields, relationships, constraints, enums, and core contracts only with migration and affected workflow review; preserve data compatibility and sensitive-field boundaries.

Required review/dependencies: Role 4 reviews migration and persistence impact; the relevant Backend and Frontend owners review workflow compatibility.

Role 4 - Database Migrations, Repositories and Reliability Engineer

Shadow segment: Database Exact file load: 22

Own migration history, sessions, repositories, the single comprehensive seed, and database reliability.

Deep responsibility scope

- Alembic configuration and revision history for reproducible schema upgrades.
- Session lifecycle, repository abstractions, transaction-safe data access, and database initialization.
- The only seed file, including RBAC metadata and all deterministic academic, finance, library, and learning data.

Files owned by this role

006. backend/alembic.ini

Purpose: Configures Alembic script paths, logging, and migration execution defaults.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

007. backend/alembic/env.py

Purpose: Connects Alembic to application metadata and the configured MySQL database for online or offline migrations.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

008. backend/alembic/script.py.mako

Purpose: Defines the template used when a new Alembic revision is generated.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

009. backend/alembic/versions/0001_initial_schema.py

Purpose: Records the reversible database schema change for 0001 initial schema.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

010. backend/alembic/versions/0002_academic_structure.py

Purpose: Records the reversible database schema change for 0002 academic structure.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

011. backend/alembic/versions/0003_section_scoped_timetables.py

Purpose: Records the reversible database schema change for 0003 section scoped timetables.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

012. backend/alembic/versions/0004_null_safe_academic_scopes.py

Purpose: Records the reversible database schema change for 0004 null safe academic scopes.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

013. backend/alembic/versions/0005_canonical_hierarchy_and_student_sequences.py

Purpose: Records the reversible database schema change for 0005 canonical hierarchy and student sequences.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

014. backend/alembic/versions/2c8000b87f5f_cascading_deletes.py

Purpose: Records the reversible database schema change for 2c8000b87f5f cascading deletes.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

015. backend/alembic/versions/444e3998ae8d_expand_library_entities.py

Purpose: Records the reversible database schema change for 444e3998ae8d expand library entities.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

016. backend/alembic/versions/812be78ed321_add_hod_id_to_branch_manual.py

Purpose: Records the reversible database schema change for 812be78ed321 add hod id to branch manual.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

017. backend/alembic/versions/86ea4d69a162_revamp_finance_models.py

Purpose: Records the reversible database schema change for 86ea4d69a162 revamp finance models.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

018. backend/alembic/versions/__init__.py

Purpose: Records the reversible database schema change for init.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

019. backend/alembic/versions/cc9cb668516c_add_versioning_support_to_timetables.py

Purpose: Records the reversible database schema change for cc9cb668516c add versioning support to timetables.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

020. backend/alembic/versions/e13d8371e6e6_upgrade_library_models.py

Purpose: Records the reversible database schema change for e13d8371e6e6 upgrade library models.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

021. backend/alembic/versions/f437010c7643_add_student_assignments.py

Purpose: Records the reversible database schema change for f437010c7643 add student assignments.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

034. backend/app/database/__init__.py

Purpose: Implements backend behavior for init.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

036. backend/app/database/seed.py

Purpose: The single authoritative seed module. It reconciles access metadata and creates the complete deterministic campus dataset across every feature.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

037. backend/app/database/session.py

Purpose: Creates the shared SQLAlchemy engine and session factory and initializes registered tables.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

057. backend/app/repositories/__init__.py

Purpose: Encapsulates database queries and persistence operations for init.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

058. backend/app/repositories/base.py

Purpose: Encapsulates database queries and persistence operations for base.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

059. backend/app/repositories/concrete.py

Purpose: Encapsulates database queries and persistence operations for concrete.

Feature contribution: Contributes to reproducible schema evolution, reliable queries, and one complete source of demonstration data.

What this role may edit/change: May change migrations, sessions, repositories, or seed behavior; rehearse upgrades, use backups for destructive work, and verify counts, constraints, and rollback.

Required review/dependencies: Role 3 reviews data meaning; every affected Backend and Frontend owner reviews migration, repository, or seed impact.

8. Backend shadow segment

This segment contains 67 files across 3 accountable roles.

Role	Title	Files
5	Backend Platform, Security and Cross-Domain API Engineer	22
6	Academic, People and Learning Backend Engineer	22
7	Finance, Library and Operational Backend Engineer	23

Role 5 - Backend Platform, Security and Cross-Domain API Engineer

Shadow segment: Backend Exact file load: 22

Own application assembly, authentication, authorization, shared API behavior, dashboards, notifications, middleware, and error handling.

Deep responsibility scope

- FastAPI composition, lifecycle, dependency injection, exception mapping, CORS, audit middleware, and routing.
- CAPTCHA, password hashing, JWT cookies, current-user resolution, permissions, and negative-access behavior.
- Shared service foundations, dashboard aggregation, notifications, and public API contract consistency.

Files owned by this role

024. backend/app/core/__init__.py

Purpose: Provides shared backend platform behavior for init.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

025. backend/app/core/config.py

Purpose: Provides shared backend platform behavior for config.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

026. backend/app/core/dependencies.py

Purpose: Provides shared backend platform behavior for dependencies.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

028. backend/app/core/exception_handlers.py

Purpose: Provides shared backend platform behavior for exception handlers.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

029. backend/app/core/exceptions.py

Purpose: Provides shared backend platform behavior for exceptions.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

031. backend/app/core/permissions.py

Purpose: Provides shared backend platform behavior for permissions.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

033. backend/app/core/security.py

Purpose: Provides shared backend platform behavior for security.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

038. backend/app/main.py

Purpose: Builds the FastAPI application, runs startup initialization, registers middleware and routers, and serves the production frontend when available.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

039. backend/app/middleware/__init__.py

Purpose: Provides request and response middleware behavior for init.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

040. backend/app/middleware/audit_middleware.py

Purpose: Provides request and response middleware behavior for audit middleware.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

060. backend/app/routers/__init__.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for init.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

064. backend/app/routers/auth.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for auth.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

065. backend/app/routers/captcha.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for captcha.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

066. backend/app/routers/dashboard.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for dashboard.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

068. backend/app/routers/notifications.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for notifications.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

073. backend/app/schemas/auth.py

Purpose: Defines Pydantic request and response contracts for auth.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

074. backend/app/schemas/common.py

Purpose: Defines Pydantic request and response contracts for common.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

076. backend/app/schemas/dashboard.py

Purpose: Defines Pydantic request and response contracts for dashboard.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

084. backend/app/schemas/user.py

Purpose: Defines Pydantic request and response contracts for user.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

088. backend/app/services/auth_service.py

Purpose: Encapsulates reusable business rules and orchestration for auth service.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

089. backend/app/services/base.py

Purpose: Encapsulates reusable business rules and orchestration for base.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

091. backend/app/services/dashboard_service.py

Purpose: Encapsulates reusable business rules and orchestration for dashboard service.

Feature contribution: Contributes to secure application-wide API behavior, access control, observability, and shared platform consistency.

What this role may edit/change: May change platform endpoints, dependencies, permissions, tokens, middleware, and error mapping; verify authentication, negative access, audit, and public contract behavior.

Required review/dependencies: Relevant Frontend and Database owners review public contracts, data access, and permission boundaries.

Role 6 - Academic, People and Learning Backend Engineer

Shadow segment: Backend **Exact file load:** 22

Own academic hierarchy, people records, faculty portfolios, attendance, marks, assignments, timetables, exports, and learning workflows.

Deep responsibility scope

- Department, course, branch, curriculum, section, student, faculty, subject, and allocation workflows.
- Attendance, marks, assignments, submissions, timetable behavior, and role-scoped faculty portfolios.
- Serialization, CSV/PDF reporting, academic documentation, role access guidance, and presentation material.

Files owned by this role

032. backend/app/core/portfolio.py

Purpose: Provides shared backend platform behavior for portfolio.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

061. backend/app/routers/academic.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for academic.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

062. backend/app/routers/api_routes.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for api routes.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

063. backend/app/routers/assignments.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for assignments.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

071. backend/app/schemas/assignment.py

Purpose: Defines Pydantic request and response contracts for assignment.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

080. backend/app/schemas/marks.py

Purpose: Defines Pydantic request and response contracts for marks.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

082. backend/app/schemas/subject.py

Purpose: Defines Pydantic request and response contracts for subject.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

083. backend/app/schemas/teacher.py

Purpose: Defines Pydantic request and response contracts for teacher.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

085. backend/app/services/__init__.py

Purpose: Encapsulates reusable business rules and orchestration for init.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

086. backend/app/services/academic_service.py

Purpose: Encapsulates reusable business rules and orchestration for academic service.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

087. backend/app/services/assignment_service.py

Purpose: Encapsulates reusable business rules and orchestration for assignment service.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

090. backend/app/services/crud_services.py

Purpose: Encapsulates reusable business rules and orchestration for crud services.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

095. backend/app/templates/email/student_welcome.html

Purpose: Defines the user-facing HTML email for student welcome.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

097. backend/app/utils/__init__.py

Purpose: Provides reusable backend utility behavior for init.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

098. backend/app/utils/csv_export.py

Purpose: Provides reusable backend utility behavior for csv export.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

099. backend/app/utils/pdf_export.py

Purpose: Provides reusable backend utility behavior for pdf export.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

100. backend/app/utils/report_factory.py

Purpose: Provides reusable backend utility behavior for report factory.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

101. backend/app/utils/serializer.py

Purpose: Provides reusable backend utility behavior for serializer.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

106. docs/academic-architecture.md

Purpose: Explains academic hierarchy, subject classification, curriculum resolution, locking, and migration behavior.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

107. docs/curriculum-guide.md

Purpose: Explains curriculum, versions, semesters, subjects, sections, and batch behavior in beginner-friendly language.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

111. docs/presentation-brief.md

Purpose: Provides the SME presentation narrative, architecture emphasis, visual direction, and slide content.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

112. docs/role-access.md

Purpose: Documents which sidebar and feature areas are visible to each application role.

Feature contribution: Contributes to correct academic, people, learning, timetable, and reporting workflows.

What this role may edit/change: May change academic and learning rules, workflow state, reporting, and serialization; manually verify admin, HOD, faculty, and student scope.

Required review/dependencies: Role 2, Role 3, Role 4, and Role 5 review user experience, data meaning, persistence, and scope enforcement.

Role 7 - Finance, Library and Operational Backend Engineer

Shadow segment: Backend Exact file load: 23

Own finance, library, email, runtime configuration, supported operations, dependencies, deployment, and operational documentation.

Deep responsibility scope

- Fee structures, invoices, payments, expenses, salaries, catalogue, circulation, reservations, fines, and email.
- Backend runtime, environment template, logging, event dispatch, uploads, containers, and dependency management.
- The small supported operations CLI plus database, execution, and deployment guidance for recoverable operation.

Files owned by this role

001. .gitignore

Purpose: Keeps dependencies, secrets, caches, builds, local databases, uploads, and logs out of source control.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

003. backend/.dockerignore

Purpose: Keeps local dependencies, caches, and generated output out of container build contexts.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

004. backend/.env.example

Purpose: Supports the project behavior represented by .env.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

005. backend/Dockerfile

Purpose: Builds the service into a repeatable production container image.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

022. backend/app/__init__.py

Purpose: Implements backend behavior for init.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

023. backend/app/cli.py

Purpose: Provides the three supported database operations: initialize system data, load the complete demo dataset, or clean demo data while preserving the administrator.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

027. backend/app/core/events.py

Purpose: Provides shared backend platform behavior for events.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

030. backend/app/core/logging_config.py

Purpose: Provides shared backend platform behavior for logging config.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

067. backend/app/routers/finance.py

Purpose: Publishes FastAPI endpoints and binds permissions, validation, services, and serialization for finance.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

069. backend/app/schemas/__init__.py

Purpose: Defines Pydantic request and response contracts for init.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

078. backend/app/schemas/finance.py

Purpose: Defines Pydantic request and response contracts for finance.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

079. backend/app/schemas/library.py

Purpose: Defines Pydantic request and response contracts for library.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

092. backend/app/services/email_service.py

Purpose: Encapsulates reusable business rules and orchestration for email service.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

093. backend/app/services/fee_strategies.py

Purpose: Encapsulates reusable business rules and orchestration for fee strategies.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

094. backend/app/services/maintenance_service.py

Purpose: Encapsulates reusable business rules and orchestration for maintenance service.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

096. backend/app/templates/email/teacher_welcome.html

Purpose: Defines the user-facing HTML email for teacher welcome.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

102. backend/requirements.txt

Purpose: Supports the project behavior represented by requirements.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

103. backend/run.py

Purpose: Starts the FastAPI application with Uvicorn using the configured host, port, reload, and log level.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

104. backend/uploads/.gitkeep

Purpose: Supports the project behavior represented by .gitkeep.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

105. docker-compose.yml

Purpose: Defines the MySQL, FastAPI, and Nginx/React services, health dependency, ports, and persistent volumes.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

108. docs/database-setup.md

Purpose: Explains fresh initialization, upgrades, seeding, backup, restore, verification, and MySQL troubleshooting.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

109. docs/deployment.md

Purpose: Describes production topology, security, persistence, migration, monitoring, and rollback expectations.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

110. docs/execution-guide.md

Purpose: Provides the end-to-end Docker and native development procedure for freshers.

Feature contribution: Contributes to correct finance/library behavior and simple, recoverable runtime operation.

What this role may edit/change: May change finance, library, email, dependencies, runtime, and operational guidance; verify transactions, reconciliation, permissions, startup, and recovery.

Required review/dependencies: Role 2 reviews user impact; Roles 3 and 4 review persisted state; Role 5 reviews security and audit behavior.

9. Complete 155-file coverage index

This compact audit repeats the primary assignment for every project-owned path. The number matches the detailed card. Every included file appears exactly once.

No.	File	Shadow segment	Primary role
1	.gitignore	Backend	Role 7
2	README.md	Frontend	Role 2
3	backend/.dockerignore	Backend	Role 7
4	backend/.env.example	Backend	Role 7
5	backend/Dockerfile	Backend	Role 7
6	backend/alembic.ini	Database	Role 4
7	backend/alembic/env.py	Database	Role 4
8	backend/alembic/script.py.mako	Database	Role 4
9	backend/alembic/versions/0001_initial_schema.py	Database	Role 4
10	backend/alembic/versions/0002_academic_structure.py	Database	Role 4
11	backend/alembic/versions/0003_section_scoped_timetables.py	Database	Role 4
12	backend/alembic/versions/0004_null_safe_academic_scopes.py	Database	Role 4
13	backend/alembic/versions/0005_canonical_hierarchy_and_student_sequences.py	Database	Role 4
14	backend/alembic/versions/2c8000b87f5f_cascading_deletes.py	Database	Role 4
15	backend/alembic/versions/444e3998ae8d_expand_library_entities.py	Database	Role 4
16	backend/alembic/versions/812be78ed321_add_hod_id_to_branch_manual.py	Database	Role 4
17	backend/alembic/versions/86ea4d69a162_revamp_finance_models.py	Database	Role 4
18	backend/alembic/versions/_init_.py	Database	Role 4
19	backend/alembic/versions/cc9cb668516c_add_versioning_support_to_timetables.py	Database	Role 4
20	backend/alembic/versions/e13d8371e6e6_upgrade_library_models.py	Database	Role 4
21	backend/alembic/versions/f437010c7643_add_student_assignments.py	Database	Role 4
22	backend/app/_init_.py	Backend	Role 7
23	backend/app/cli.py	Backend	Role 7
24	backend/app/core/_init_.py	Backend	Role 5
25	backend/app/core/config.py	Backend	Role 5
26	backend/app/core/dependencies.py	Backend	Role 5
27	backend/app/core/events.py	Backend	Role 7
28	backend/app/core/exception_handlers.py	Backend	Role 5
29	backend/app/core/exceptions.py	Backend	Role 5
30	backend/app/core/logging_config.py	Backend	Role 7
31	backend/app/core/permissions.py	Backend	Role 5
32	backend/app/core/portfolio.py	Backend	Role 6
33	backend/app/core/security.py	Backend	Role 5
34	backend/app/database/_init_.py	Database	Role 4
35	backend/app/database/base.py	Database	Role 3
36	backend/app/database/seed.py	Database	Role 4
37	backend/app/database/session.py	Database	Role 4
38	backend/app/main.py	Backend	Role 5
39	backend/app/middleware/_init_.py	Backend	Role 5
40	backend/app/middleware/audit_middleware.py	Backend	Role 5
41	backend/app/models/_init_.py	Database	Role 3
42	backend/app/models/academic.py	Database	Role 3

No.	File	Shadow segment	Primary role
43	backend/app/models/attendance.py	Database	Role 3
44	backend/app/models/audit_log.py	Database	Role 3
45	backend/app/models/course.py	Database	Role 3
46	backend/app/models/department.py	Database	Role 3
47	backend/app/models/finance.py	Database	Role 3
48	backend/app/models/library.py	Database	Role 3
49	backend/app/models/marks.py	Database	Role 3
50	backend/app/models/notification.py	Database	Role 3
51	backend/app/models/student.py	Database	Role 3
52	backend/app/models/subject.py	Database	Role 3
53	backend/app/models/teacher.py	Database	Role 3
54	backend/app/models/timetable.py	Database	Role 3
55	backend/app/models/timetable_grid.py	Database	Role 3
56	backend/app/models/user.py	Database	Role 3
57	backend/app/repositories/__init__.py	Database	Role 4
58	backend/app/repositories/base.py	Database	Role 4
59	backend/app/repositories/concrete.py	Database	Role 4
60	backend/app/routers/__init__.py	Backend	Role 5
61	backend/app/routers/academic.py	Backend	Role 6
62	backend/app/routers/api_routes.py	Backend	Role 6
63	backend/app/routers/assignments.py	Backend	Role 6
64	backend/app/routers/auth.py	Backend	Role 5
65	backend/app/routers/captcha.py	Backend	Role 5
66	backend/app/routers/dashboard.py	Backend	Role 5
67	backend/app/routers/finance.py	Backend	Role 7
68	backend/app/routers/notifications.py	Backend	Role 5
69	backend/app/schemas/__init__.py	Backend	Role 7
70	backend/app/schemas/academic.py	Database	Role 3
71	backend/app/schemas/assignment.py	Backend	Role 6
72	backend/app/schemas/attendance.py	Database	Role 3
73	backend/app/schemas/auth.py	Backend	Role 5
74	backend/app/schemas/common.py	Backend	Role 5
75	backend/app/schemas/course.py	Database	Role 3
76	backend/app/schemas/dashboard.py	Backend	Role 5
77	backend/app/schemas/department.py	Database	Role 3
78	backend/app/schemas/finance.py	Backend	Role 7
79	backend/app/schemas/library.py	Backend	Role 7
80	backend/app/schemas/marks.py	Backend	Role 6
81	backend/app/schemas/student.py	Database	Role 3
82	backend/app/schemas/subject.py	Backend	Role 6
83	backend/app/schemas/teacher.py	Backend	Role 6
84	backend/app/schemas/user.py	Backend	Role 5
85	backend/app/services/__init__.py	Backend	Role 6
86	backend/app/services/academic_service.py	Backend	Role 6
87	backend/app/services/assignment_service.py	Backend	Role 6
88	backend/app/services/auth_service.py	Backend	Role 5

No.	File	Shadow segment	Primary role
89	backend/app/services/base.py	Backend	Role 5
90	backend/app/services/crud_services.py	Backend	Role 6
91	backend/app/services/dashboard_service.py	Backend	Role 5
92	backend/app/services/email_service.py	Backend	Role 7
93	backend/app/services/fee_strategies.py	Backend	Role 7
94	backend/app/services/maintenance_service.py	Backend	Role 7
95	backend/app/templates/email/student_welcome.html	Backend	Role 6
96	backend/app/templates/email/teacher_welcome.html	Backend	Role 7
97	backend/app/utlis/__init__.py	Backend	Role 6
98	backend/app/utlis/csv_export.py	Backend	Role 6
99	backend/app/utlis/pdf_export.py	Backend	Role 6
100	backend/app/utlis/report_factory.py	Backend	Role 6
101	backend/app/utlis/serializer.py	Backend	Role 6
102	backend/requirements.txt	Backend	Role 7
103	backend/run.py	Backend	Role 7
104	backend/uploads/.gitkeep	Backend	Role 7
105	docker-compose.yml	Backend	Role 7
106	docs/academic-architecture.md	Backend	Role 6
107	docs/curriculum-guide.md	Backend	Role 6
108	docs/database-setup.md	Backend	Role 7
109	docs/deployment.md	Backend	Role 7
110	docs/execution-guide.md	Backend	Role 7
111	docs/presentation-brief.md	Backend	Role 6
112	docs/role-access.md	Backend	Role 6
113	frontend/.dockerignore	Frontend	Role 1
114	frontend/Dockerfile	Frontend	Role 1
115	frontend/eslint.config.js	Frontend	Role 1
116	frontend/index.html	Frontend	Role 1
117	frontend/nginx.conf	Frontend	Role 1
118	frontend/package-lock.json	Frontend	Role 1
119	frontend/package.json	Frontend	Role 1
120	frontend/public/institute.png	Frontend	Role 1
121	frontend/src/App.jsx	Frontend	Role 1
122	frontend/src/components/layout/AppShell.jsx	Frontend	Role 1
123	frontend/src/components/layout/PermissionGate.jsx	Frontend	Role 1
124	frontend/src/components/ui/Button.jsx	Frontend	Role 1
125	frontend/src/components/ui/Card.jsx	Frontend	Role 1
126	frontend/src/components/ui/ConfirmDialog.jsx	Frontend	Role 1
127	frontend/src/components/ui/PageHeader.jsx	Frontend	Role 1
128	frontend/src/components/ui/PeopleDirectory.jsx	Frontend	Role 2
129	frontend/src/components/ui/ProfileDetailModal.jsx	Frontend	Role 2
130	frontend/src/components/ui/RecordFormModal.jsx	Frontend	Role 2
131	frontend/src/components/ui/StatCard.jsx	Frontend	Role 1
132	frontend/src/config/resourceForms.js	Frontend	Role 2
133	frontend/src/config/resources.jsx	Frontend	Role 2
134	frontend/src/config/roleAccess.js	Frontend	Role 2

No.	File	Shadow segment	Primary role
135	frontend/src/lib/api.js	Frontend	Role 1
136	frontend/src/main.jsx	Frontend	Role 1
137	frontend/src/pages/AssignmentsPage.css	Frontend	Role 2
138	frontend/src/pages/AssignmentsPage.jsx	Frontend	Role 2
139	frontend/src/pages/AttendancePage.css	Frontend	Role 2
140	frontend/src/pages/AttendancePage.jsx	Frontend	Role 2
141	frontend/src/pages/DashboardPage.jsx	Frontend	Role 2
142	frontend/src/pages/FeeCollectionPage.css	Frontend	Role 2
143	frontend/src/pages/FeeCollectionPage.jsx	Frontend	Role 2
144	frontend/src/pages/HodManagementPage.jsx	Frontend	Role 2
145	frontend/src/pages/HodPanelPage.jsx	Frontend	Role 2
146	frontend/src/pages/LoginPage.jsx	Frontend	Role 2
147	frontend/src/pages/NotificationsPage.jsx	Frontend	Role 2
148	frontend/src/pages/ProfilePage.jsx	Frontend	Role 2
149	frontend/src/pages/ResourcePage.jsx	Frontend	Role 2
150	frontend/src/pages/TimetablePage.jsx	Frontend	Role 2
151	frontend/src/state/AuthContext.jsx	Frontend	Role 1
152	frontend/src/styles/dashboard-widgets.css	Frontend	Role 1
153	frontend/src/styles/index.css	Frontend	Role 1
154	frontend/src/styles/people-directory.css	Frontend	Role 2
155	frontend/vite.config.js	Frontend	Role 1

10. Definition of done

- Role 1: frontend lint/build passes; shared UI remains responsive and safe under loading, error, and session failure.
- Role 2: every workflow handles validation, loading, empty, success, and failure states; browser visibility matches backend authority.
- Role 3: model and contract changes preserve meaning, constraints, compatibility, and explicit sensitive-field decisions.
- Role 4: migration and seed changes are rehearsed; backup/restore impact, counts, constraints, and rollback are understood.
- Role 5: authentication, authorization, negative access, error mapping, audit, and public API behavior are manually verified.
- Role 6: academic scope and learning transitions are correct for admin, HOD, faculty, and student contexts.
- Role 7: finance and circulation state is correct and auditable; runtime and maintenance changes are recoverable and documented.
- All roles: no live secret, personal data, credential, unsafe generated log, cache, or build output is committed.
- Project rule: new demo data is added only to backend/app/database/seed.py; no temporary seed or reseed scripts are reintroduced.
- Current verification baseline: 151 backend routes load, Alembic current equals head, frontend lint/build passes, and authenticated smoke calls succeed across all major feature areas.

Presentation takeaway: the folder is simpler without flattening the architecture. Freshers can follow one path from router to schema, service, repository, model, and database - and one seed file demonstrates the entire campus.